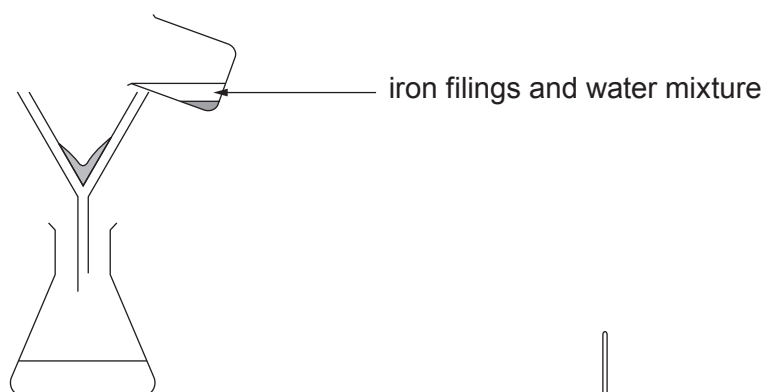


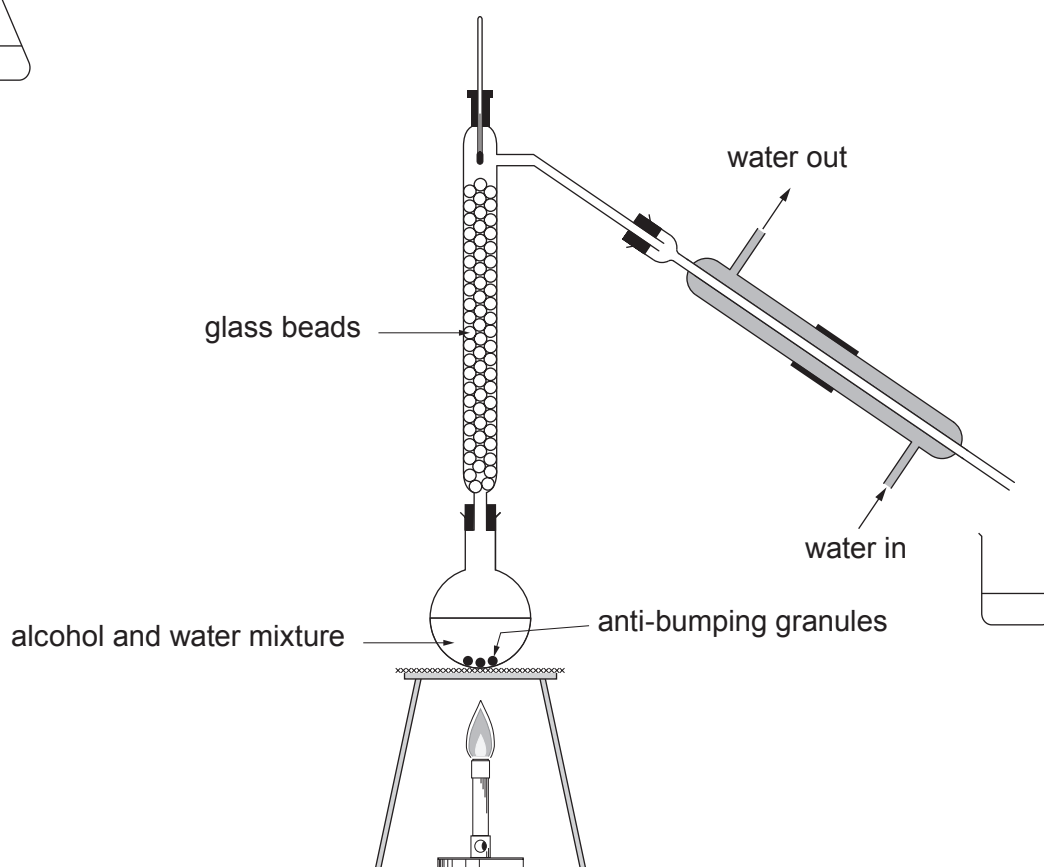
Answer **all** questions.

1. (a) The diagrams show three methods used to separate mixtures.

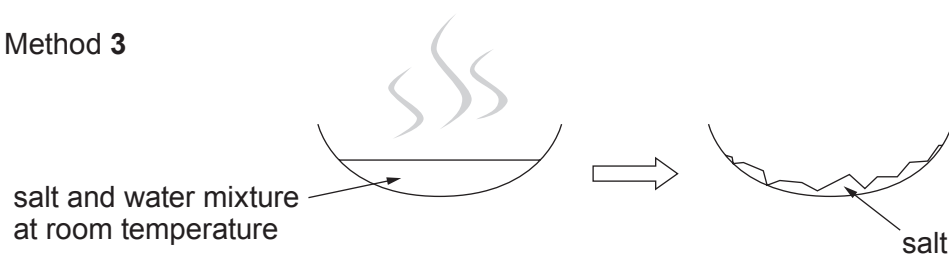
Method 1



Method 2



Method 3



- (i) Underline the property of iron filings which allows method **1** to be used to separate the mixture. [1]

magnetic

insoluble in water

silvery grey colour

high melting point

more dense than water

- (ii) Tick (✓) the box that shows the reason why method **2** can be used to separate alcohol and water. [1]

alcohol and water both boil when heated

☐

alcohol and water have different boiling points

☐

alcohol and water have the same boiling point

☐

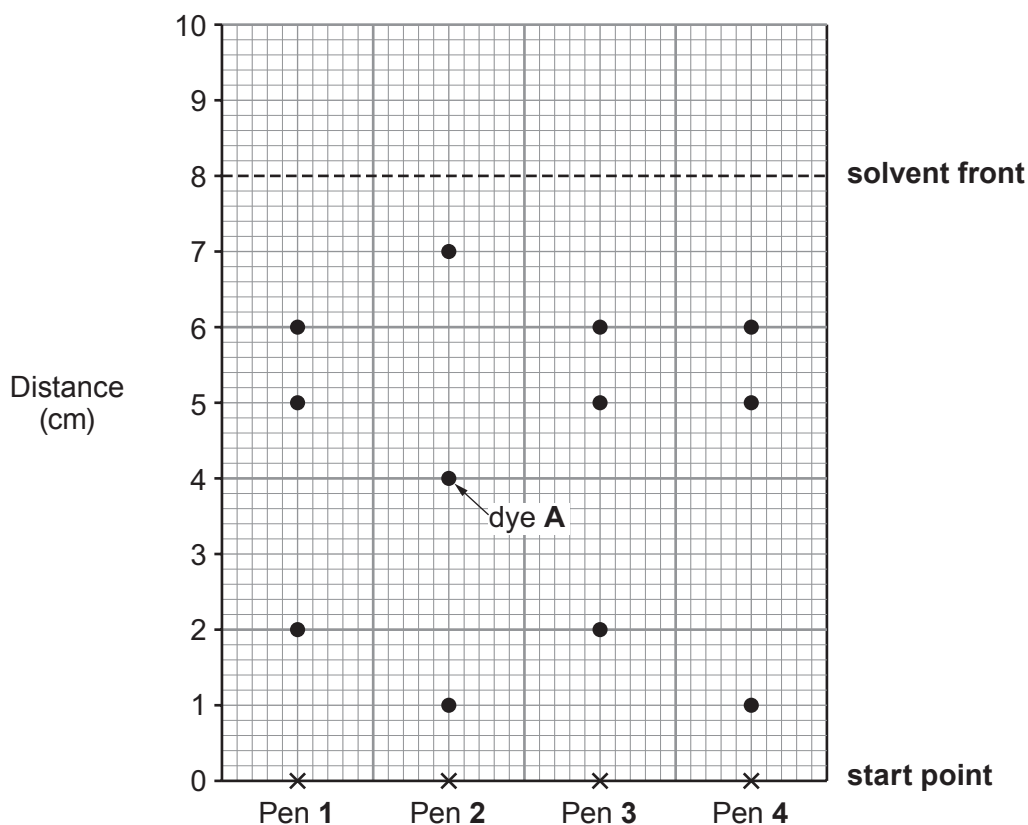
alcohol and water are both liquids

☐

- (iii) Name the process of removing water in method **3**. [1]

.....

- (b) A group of students were asked to investigate the dyes in four different water soluble blue pens, **1**, **2**, **3** and **4**. The results are shown below.



- (i) Which **two** pens contain the same three dyes? [1]
Pens and
- (ii) Which pen contains the **most** soluble dye? [1]
Pen
- (iii) The R_f value of a substance can be used to identify that substance.
The R_f value is given by the formula:

$$R_f = \frac{\text{distance moved by the substance}}{\text{distance moved by the solvent front}}$$

Calculate the R_f value for dye **A**. [2]

$R_f =$

- (iv) The box contains methods for separating mixtures.

distillation**chromatography****filtration**

Complete the sentence using a method from the box.

The method used to separate the coloured dyes in the blue pens is called

.....

[1]

8